

Quick Note on Angular Momentum Addition

PHYS 312

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1 Angular Momentum from two Subspaces

- Angular Momentum operator $\hat{\mathbf{J}}$ (I will write just $\mathbf{J}$) is the **Rotation Generator** of any ket $|\Psi\rangle$. By generator we mean that using $\mathbf{J}$ we can write an **Rotation Operator**

$$\mathcal{R}(\hat{n}, d\phi) = 1 - i \left(\frac{\mathbf{J} \cdot \hat{n}}{\hbar} \right) d\phi \quad (1)$$

Simplification of the above garbage can give us something more usable. Invoke $\hat{n} = \hat{z} = \mathbf{e}_z$ then

$$\mathcal{R}_z(d\phi) = 1 - i \left(\frac{J_z}{\hbar} \right) d\phi$$

Finite rotation can be achieved by continuously doing infinitesimal rotations

$$\underbrace{\mathcal{R}_z(d\phi)\mathcal{R}_z(d\phi)\dots\mathcal{R}_z(d\phi)}_{n \text{ times}} = R_z(\phi) = \left[1 - i \left(\frac{J_z}{\hbar} \right) d\phi \right]^n = \left[1 - i \left(\frac{J_z}{\hbar} \right) \left(\frac{\phi}{n} \right) \right]^n$$

$$\implies R_z(\phi) = \exp \left(-i \frac{J_z \phi}{\hbar} \right) \quad (2)$$

- Consider the two angular momentum operators $\mathbf{J}_1$ and $\mathbf{J}_2$ that are member of two different subspaces. They satisfy the following **Angular Momentum Algebra** as usual

$$[J_{1i}, J_{1j}] = i\hbar \varepsilon_{ijk} J_{1k} \quad (3)$$

$$[J_{2i}, J_{2j}] = i\hbar \varepsilon_{ijk} J_{2k} \quad (4)$$

- But because the two angular momentum are from different subspaces we have

$$[J_{1m}, J_{2n}] = 0$$

2 Addition of the two Angular Momentum

- The infinitesimal rotation operator that affects both subspaces 1 and 2 can be written as the tensor product where 1_1 and 1_2 represent the Identity Operators of subspace 1 and 2 respectively.

$$\mathcal{R}_{1 \text{ and } 2} = \mathcal{R}_1 \otimes \mathcal{R}_2 = \left(1_1 - i \left(\frac{\mathbf{J}_1 \cdot \hat{n}}{\hbar} \right) d\phi \right) \otimes \left(1_2 - i \left(\frac{\mathbf{J}_2 \cdot \hat{n}}{\hbar} \right) d\phi \right)$$

$$\mathcal{R}_{1 \text{ and } 2} = 1 - i \frac{(\mathbf{J}_1 \otimes 1_2 + 1_1 \otimes \mathbf{J}_2) \cdot \mathbf{n} \cdot \phi}{\hbar}$$

- Defining **Total Angular Momentum** of the two subspace as

$$\mathbf{J} = \mathbf{J}_1 \otimes 1_2 + 1_1 \otimes \mathbf{J}_2 \equiv \mathbf{J}_1 + \mathbf{J}_2$$

It can be proved (as an exercise or just taking it for granted) that the total angular momentum satisfies the required Algebra

$$[J_i, J_j] = i\hbar \varepsilon_{ijk} J_k$$

This gives us the Green Signal to continue with the mathematics treating the total angular momentum $\mathbf{J}$ as an ordinary angular momentum.

- $\mathbf{J}$ is to be treated as a Generator for the Entire System. We can follow the definition of $\mathbf{J}^2 = J_x J_x + J_y J_y + J_z J_z$ for the whole system and hence have the following relations that we have previously seen like

$$[\mathbf{J}^2, J_z] = 0$$

Here the J_z will represent the *component of the angular momentum* for the whole system instead of a single subspace.

- **Warning:** $[\mathbf{J}^2, J_{1z}] \neq 0$, same for subspace 2.
- $\mathbf{J}^2$ can be represented in terms of $\mathbf{J}_1$ and $\mathbf{J}_2$.

$$\mathbf{J}^2 = \mathbf{J}_1^2 + \mathbf{J}_2^2 + 2J_{1z}J_{2z} + J_{1+}J_{2-} + J_{1-}J_{2+} \quad (5)$$

This can be solved by taking square of $\mathbf{J}_1 + \mathbf{J}_2$ and then replacing $\mathbf{J}_1 \cdot \mathbf{J}_2$ with J_{1z}, J_{2z} and Ladder Operators instead of J_{1x}, J_{2x} , etc.

- Above says

$$[\mathbf{J}^2, \mathbf{J}_1^2] = 0 \quad (6)$$

$$[\mathbf{J}^2, \mathbf{J}_2^2] = 0 \quad (7)$$

Eigenkets of the Individual Operators

- For completeness let's quickly review the eigenkets for individual operators $\mathbf{J}_1^2, \mathbf{J}_2^2$

$$\mathbf{J}_1^2 |j_1, j_2, m_1, m_2\rangle = j_1(j_1 + 1)\hbar^2 |j_1, j_2, m_1, m_2\rangle$$

$$\mathbf{J}_2^2 |j_1, j_2, m_1, m_2\rangle = j_2(j_2 + 1)\hbar^2 |j_1, j_2, m_1, m_2\rangle$$

$$J_{1z} |j_1, j_2, m_1, m_2\rangle = m_1\hbar |j_1, j_2, m_1, m_2\rangle$$

$$J_{2z} |j_1, j_2, m_1, m_2\rangle = m_2\hbar |j_1, j_2, m_1, m_2\rangle$$

Eigenkets of the Combined Operators

- Because we have found the operators, we should expect to find the Eigenkets now. $\mathbf{J}_1^2, \mathbf{J}_2^2$ commutes with $\mathbf{J}^2$ hence we can have simultaneous eigenkets such that

$$\mathbf{J}_1^2 |j_1, j_2, j, m\rangle = j_1(j_1 + 1)\hbar^2 |j_1, j_2, j, m\rangle \quad (8)$$

$$\mathbf{J}_2^2 |j_1, j_2, j, m\rangle = j_2(j_2 + 1)\hbar^2 |j_1, j_2, j, m\rangle \quad (9)$$

$$\mathbf{J}^2 |j_1, j_2, j, m\rangle = j(j + 1)\hbar^2 |j_1, j_2, j, m\rangle \quad (10)$$

$$J_z |j_1, j_2, j, m\rangle = m\hbar |j_1, j_2, j, m\rangle \quad (11)$$

Using $||\Psi\rangle\rangle$ instead of $|\Psi\rangle$ is going to be our resort to differentiate between combined eigenstates and individual eigenstates.

Transformation between Individual and Combined

- We know that Individual Eigenkets and Combined Eigenkets are not the same because $[\mathbf{J}^2, J_{1z}] \neq 0$ don't commute. So they are different. There exists a transformation between them, which we can do by the following: Create a projection operator using the individual eigenkets

$$\hat{I} = \sum_{m_1} \sum_{m_2} |j_1, j_2, m_1, m_2\rangle \langle j_1, j_2, m_1, m_2|$$

A combined ket $||\mathbf{j}_1, \mathbf{j}_2, \mathbf{j}, \mathbf{m}\rangle\rangle$ in this representation would be

$$||j_1, j_2, j, m\rangle\rangle = \sum_{m_1} \sum_{m_2} |j_1, j_2, m_1, m_2\rangle \langle j_1, j_2, m_1, m_2 ||\mathbf{j}_1, \mathbf{j}_2, \mathbf{j}, \mathbf{m}\rangle\rangle$$

The term $\langle j_1, j_2, m_1, m_2 ||\mathbf{j}_1, \mathbf{j}_2, \mathbf{j}, \mathbf{m}\rangle\rangle$ is called the Clebsh-Gordan Coefficient.